

SW Engineering CSC648/848

Specced

Milestone 1

User Stories & High-Level Requirements

Section 6 | Team 5

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Student Name	Role
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1. Executive Summary:

The Problem

Writing PRDs is a laborious task for Project Managers. They have to synthesize research, align stakeholders, and structure a clear, complete document before engineering can even begin. This work eats into the time PMs could otherwise dedicate to strategy, customer conversations, and roadmap decisions. But PRDs are integral to how product work gets prioritized and handed off — it's that essential they are created well to keep everyone on track. The result is that every team downstream waits on the PRD, and maintaining high-quality PRDs takes hours for PMs.

The Solution

Our solution is a PRD generator and classifier, an AI-powered workflow that will let Product Managers write and assign PRDs in a fraction of the time it takes today.

An AI assistant helps a PM draft a complete PRD from a short brief or set of notes, producing a structured, high-quality first draft in minutes instead of hours. It also helps with the research behind the PRD, pulling from real, verifiable sources rather than fabricating information. Once the generator is in the hands of PMs, we plan to add a second AI assistant that reads the finished PRD and suggests not just the owning team, but priority, dependencies, and required approvals. Every suggestion has to go through human review. The AI proposes, the PM decides whether to accept it, and only then is the PRD assigned in-product. Those accept/reject decisions feed back into the classifier, so it keeps improving from real PM judgment. From there, PMs and engineers will be able to exchange notes on the PRD without leaving the tool — turning a slow, manual process into a fast, guided one from draft to assignment.

Our Roadmap

We're taking a deliberately staged approach.

- ★ **Phase 1** (current focus): The PRD Generator. Ship the AI drafting assistant on its own, so PMs get immediate time savings and we can validate output quality with real teams before adding routing logic.
- ★ **Phase 2**: The Classifier. Layer in the team, priority, dependency, and approval-suggestion assistant, with every suggestion subject to PM review and acceptance. PM accept/reject choices become the training signal that improves the classifier over time.
- ★ **Phase 3**: Collaboration layer. Expand PM/engineer note-sharing and feedback loops on assigned PRDs.

Scoping Phase 1 tightly lets us prove the core value proposition with a small, shippable release.

2. Personas and User Stories

These personas describe the people who may use our PRD Generator.

2.1 Personas

Persona	Behavior / Skills	Pain Points	Primary Goal
P1 – Product Manager	Maya is a PM at a small software company in San Francisco. Most of her week goes to planning features with designers and engineers. On weekends, she is usually out hiking.	Meeting notes end up in one document, feedback in another, and the rest in Slack threads. Pulling it all together before a planning review takes hours.	Create a PRD her team can read and discuss before anyone starts building.
P2 – Startup Founder	After growing up in Oakland, Daniel started a small online store. He talks to customers almost every day and has an app idea he keeps coming back to. He plays pickup basketball with friends when he can.	Some of his ideas are in his phone's notes app. He remembers the rest from conversations. He has never written a PRD.	Explain his idea clearly to the people who might help him build it.
P3 – Product Engineer / Tech Lead	Alex is a tech lead based in Daly City. He reviews feature requests and estimates how much work they will take. He cooks when he finds the time.	Requests often say what to build but leave out who it is for or why. He ends up asking the team for the missing details.	Understand what each feature is for and where its limits are before development starts.
P4 – Student / Junior PM	At 23, Nina moved from Sacramento to San Francisco for school. She is into photography and is writing her first PRD for a class project.	She knows her idea well but gets stuck on sections like user needs and success metrics. She is not sure what belongs there.	Answer a few clear questions and finish a PRD she feels comfortable sharing with classmates.

2.2 User Stories

These stories describe what people want to do with the app and why.

ID	Activity	User Story	Priority
US1	Planning	At the start of a new project, Maya wants to put the product idea, the problem it solves, and the main goals in one place so her team knows what the PRD is about.	High
US2	Drafting	Daniel has talked about his app idea with friends but has never put it all in one document. After answering the product questions, he wants a first PRD draft he can review with them.	High
US3	Review and Editing	When Maya's team drops a feature after a meeting, she wants to update the PRD so nobody keeps working from the old plan.	High
US4	Drafting	One section still sounds wrong to Maya, even though she is happy with the rest of the PRD. She wants a new draft of just that section without losing her edits elsewhere.	Mid
US5	PRD Management	Maya works on several PRDs at once. She wants to create an account so she can sign in, save and reopen documents, sign out, and delete a PRD she no longer needs.	High
US6	Planning	Nina has to write her first PRD for class, but she is unsure what belongs under user needs or success metrics. She wants questions about users, goals, features, limits, and measures of success to guide her.	High
US7	Review and Editing	A draft mentions a customer need Daniel never described. He wants that guess pointed out so he can correct it before sharing the PRD.	Mid
US8	Review and Editing	Before building his own project, Alex wants to read its PRD to check the users, goals, requirements, and limits. That helps him spot anything he needs to clarify.	High
US9	PRD Management	Halfway through the questions, Daniel realizes he needs to ask a customer about one feature. He wants to save his answers now and finish later instead of starting over.	High
US10	Review and Editing	Maya comes back from a planning meeting unsure whether a requirement changed. She wants to look at an earlier PRD version to see what the team updated.	Mid
US11	PRD Management	A few weeks after pausing a project, Maya wants to find its PRD on her dashboard and pick up where she left off.	High
US12	PRD Management	Daniel needs feedback from a teammate who does not use the app. He wants to export a copy of the PRD so they can read it and send him comments.	Mid

3. Data Definitions

Term	Definition	Usage in This Product
User	A registered person who uses the application to create, manage, and edit PRDs.	Owns projects and PRDs and interacts with the drafting workflow.
Project	A product or software initiative for which the user wants to create a PRD.	Groups the product idea, context, and related PRD content in one place.
PRD	A Product Requirements Document describing the product problem, users, goals, requirements, scope, and related planning information.	Serves as the main document generated, reviewed, edited, and managed in the application.
Guided Interview	A structured sequence of questions used to collect information about the user's product idea.	Collects the context the AI needs before generating a PRD.
Interview Answers	The information provided by the user during the guided interview, such as the problem, target users, goals, features, and constraints.	Provides the main source of information used to generate the PRD.
PRD Section	An individual part of a PRD, such as Problem Statement, Personas, User Stories, Requirements, Risks, or Success Metrics.	Allows users to review, edit, and potentially regenerate specific parts of the PRD.
Persona	A representative category of target user with common goals, behaviors, needs, and pain points.	Provides user context for the generated PRD and helps define product requirements.
User Story	A short statement describing what a target user wants to accomplish and why.	Connects user needs to the features and functional requirements of the proposed product.
Requirement	A capability or quality that the planned product is expected to satisfy.	Used to describe both functional and non-functional expectations in the PRD.

Assumption	Information inferred when the user has not provided enough detail about a part of the product.	Makes AI-generated assumptions visible so the user can review or correct them instead of presenting them as facts.
Revision	A saved change or updated version of PRD content.	Supports continued editing, refinement, and version tracking over time.
AI Generation	The process of using an AI service to draft or refine PRD content based on user-provided information.	Assists with creating the initial PRD and improving individual sections while keeping the user responsible for final review.

4. Initial List of Functional Requirements

ID	Title	Description	Priority	User Story
R_1	User Authentication	The system will allow a user to create an account, sign in, and sign out.	High	US5
R_2	Create Project	The ability to let the user create a project and entire the project info	High	US1
R_3	Guided PRD Input	The system shall collect product context through a structured multi-step workflow covering the problem, users, features, goals, constraints, and success metrics.	High	US1, US6
R_4	Generate PRD Draft	The system shall generate a structured PRD draft from the information provided by the user.	High	US2
R_5	Edit PRD	Allowing the user to manually edit generated PRD content before saving it.	High	US3
R_6	Regenerate Section	Allows the user to request a new AI-generated version of an individual PRD section without regenerating the complete document.	Mid	US4

R_7	Save and Retrieve PRDs	The system shall save a user's PRDs and allow the user to reopen previously saved documents.	High	US5
R_8	Delete PRD	The system shall allow the owner of a PRD to delete a document they no longer need.	Low	US5
R_9	Save Interview Progress	The system shall save a user's partially completed interview answers so the user can leave and resume later without starting over.	High	US9
R_10	Flag AI Assumptions	The system shall mark any PRD content the AI inferred without user input as an assumption, so the user can review or correct it	Mid	US7
R_11	Export PRD	The system shall allow the user to export a PRD as a file that people without an account can read.	Mid	US12

5. Non-Functional Requirements

ID	Category	Requirement
NFR_1	Usability	The core PRD creation workflow shall be understandable to a first-time user without requiring technical product-management knowledge or outside help.
NFR_2	Performance	Standard dashboard and saved-document actions should respond promptly under expected classroom/demo usage; AI generation should display a clear loading state while processing.
NFR_3	Security	AI provider secrets and privileged database credentials shall not be exposed in client-side code.
NFR_4	Privacy	Authenticated users shall only be able to access their own saved projects and PRDs.
NFR_5	Compatibility	The web application shall support current desktop versions of Chrome and Firefox; additional browsers may be tested as time permits.

NFR_6	Availability	The deployed application should be accessible remotely for course review except during planned maintenance or third-party service outages.
NFR_7	Reliability	The application shall display understandable errors when AI generation, database access, or network requests fail.
NFR_8	Accessibility	Forms, controls, and navigation shall have readable labels, keyboard-accessible interaction, and sufficient visual clarity for common web usage.
NFR_9	Expected Users	The system shall support up to 50 concurrent users, the expected load for class demos and instructor review.
NFR_10	Storage	Each user shall be able to store multiple projects and PRDs. Total data shall stay within the Supabase free-tier limit.
NFR_11	Fault tolerance	If AI generation fails or times out, the user's interview answers and existing PRD content shall be kept, so the user can retry without re-entering anything. Interview progress shall be saved so a browser crash or lost connection does not force the user to start over.
NFR_12	Development	Every pull request shall be reviewed and approved by at least one other team member before merging to main.

6. Competitive Analysis

Feature	ChatPRD	Notion AI	Productboard Spark	Our PRD Generator (planned)
AI-assisted PRD drafting	Dedicated PRD generation from prompts and context	AI document drafting within a workspace	AI product specifications grounded in product context	Structured PRD generated from guided answers
Guided input	Conversational drafting and follow-up	Templates and user-written prompts	Guided dialogue; surfaces open questions	Multi-step interview on users, goals, scope and features

Editing and refinement	Targeted section edits and version comparisons	Manual editing and AI rewriting	AI-assisted spec refinement	Manual edits; section regeneration planned
Project/document management	Shared projects, document history	Pages, databases and collaboration	Product specifications within broader PM workflows	Personal dashboard and saved PRDs
External product context	Notion, Linear and GitHub integrations	Workspace content and connected tools	Customer feedback, strategy and codebase context	User-provided context; external data is future scope
Export / history	Exports and document versioning	Export and page history	Specs integrated with delivery workflows	Export and revision history are planned

Our proposed PRD Generator focuses on an accessible, structured workflow for product managers, startup founders, and students who may not know how to write a PRD. Although existing competitors already support AI drafting and refinement, our planned guided interview gathers core requirements before generation, while the dashboard keeps drafts organized without requiring a broader product-management workspace. Users will be able to edit and save generated PRDs; section-level regeneration, exports, and revision history remain planned features. Our initial version will rely on user-provided context rather than claiming the richer external integrations offered by established products.

Competitor references:

chatprd.ai

notion.com/product/ai

productboard.com/product/spark/

7. High-Level System Requirements

Component	Technology / Tool	Purpose
Frontend	React + Vite	Web user interface, guided PRD workflow, dashboard, and PRD editor.

Backend	Python + FastAPI	Server-side API, application logic, authorization checks, and AI requests.
Database	Supabase PostgreSQL	Stores users/profile-related data, projects, PRD inputs, and saved PRDs.
Authentication	Supabase Auth	Account creation, login, and authenticated user sessions.
AI	[Gemini / OpenAI / Claude - final provider TBD]	Generates structured PRD drafts and optionally regenerates selected sections.
Hosting	Vercel	Remote deployment of the web application/backend functions as supported by the project setup.
Version Control	GitHub	Branches, pull requests, reviews, merges, and milestone documents.
Supported Browsers	Chrome and Firefox (desktop)	Primary browsers targeted for M1/MVP testing.
Development Tools	Visual Studio Code, Postman	Source editing and API testing.

8. Team and Study Plan

Team Member	Role	Study / Technical Focus	One-Month Goal
Om Pandya	Team Lead, AI Master	AI integration, structured output, project coordination	Coordinate M1/M2 scope and complete a working AI generation proof of concept that returns structured PRD sections through the backend.
Aksh Patel	Frontend Lead, GitHub Co-Lead	React/Vite UX patterns, PRD editor workflow	Build the initial dashboard and multi-step PRD input flow, then connect at least one frontend flow to the backend API.

Giovannie Silva	AI Master	Prompt design, AI provider APIs, output validation	Test candidate AI providers/prompts and document a reliable structured response format for PRD generation.
Leman Yuksel	Scrum Master	Supabase/Auth/deployment support; process documentation	Maintain meeting notes/study tracking and complete one end-to-end environment setup/reproduction test using the team README.
Marco Garcia	Backend Lead	FastAPI API design, database integration	Implement the core backend project/PRD endpoints and verify read/write connectivity with Supabase.
Philip Chen	GitHub Master	Git workflow, deployment workflow, integration support	Ensure all members can use the branch/PR process and document/rehearse the deployment workflow for the current stack.
Vineela Vandanapu	Backend Co-Lead / Product Support	FastAPI, Supabase data model, product requirements	Support the initial data model and backend integration and help validate that the system data structures match the M1 definitions.

9. Checklist

Checklist Item	Status	Issue Notes (only if ISSUE)
Team found a time slot to meet outside of class.	DONE	
Scrum Master shares meeting minutes after each meeting.	DONE	

GitHub Master chosen.	DONE	
Everyone sets up their local development environment from the team GitHub repo.	DONE	
Team agreed on the listed software tools and deployment server.	ON TRACK	
Team is ready and able to use the chosen backend/frontend frameworks.	DONE	
For each major technology, a study lead and specific one-month study goal have been identified.	DONE	
Team Lead ensured all team members read, understand, and agree with the final M1 before submission.	DONE	